Generating Chest X-rays with a Convolutional VAE

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Abstract

We train a convolutional VAE on 128×128 chest X-rays of the Kaggle pneumonia dataset. A perceptual loss, residual blocks and an upsampling decoder make the ribs visible: prior samples reach FID 126.0 and IS 1.64 on the test images (real images: FID 26.4), with coherent anatomy but smoother than real X-rays. Code: `https://github.com/henrik-ra/chest-xray-vae`.

1 Method and hyperparameters

Data. We use the Kaggle chest X-ray pneumonia dataset [1]. With only 16 official validation images, we also hold out 10% of the training images: 4,695 training, 537 validation and 624 test images. The final configuration was selected on the validation images; the test images only evaluate the final model. Images are grayscale, resized to 128×128 (squeezing the median aspect ratio of 1.41) and scaled to $[0, 1]$; training images are shifted by up to 4 whole pixels.

Model and loss. The encoder has five stride-2 convolutions, each followed by GroupNorm, LeakyReLU and two residual blocks, and linear heads for a diagonal Gaussian posterior. The decoder mirrors it (with two more residual blocks at its 4×4 input), but upsamples by nearest-neighbour interpolation and a 3×3 convolution instead of transposed convolutions; the log-variance is softly bounded to ± 20. With decoder mean $\hat{x}(z)$ we minimize, in nats per image, the negative ELBO plus a perceptual term:

$$\mathcal{L} = \mathbb{E}_{q(z|x)}\left[\tfrac{1}{2\sigma^2}\|x - \hat{x}(z)\|^2 + \lambda\, d_{\mathrm{VGG}}(\hat{x}(z), x)\right] + \beta\,\mathrm{KL}\big(q(z|x)\,\|\,\mathcal{N}(0, I)\big) + \text{const.} \quad (1)$$

d_{VGG} is the squared distance of channel-normalized VGG16 features (relu1_2, relu2_2, relu3_3). We keep the checkpoint with the lowest validation $\mathcal{L}$.

Table 1: Hyperparameters of the final model (trained on an NVIDIA A30).

Latent dim.	128	Optimizer	Adam, lr $5 \cdot 10^{-4}$, cosine to $5 \cdot 10^{-6}$
Channels	32 to 512 (32.7M params)	Batch, epochs	32, 160 (best: 137)
Likelihood	Gaussian, $\sigma = 0.25$	KL weight	$\beta = 1$, 10-epoch warm-up
Percept. weight	$\lambda = 1440$	Clipping, seed	grad. norm 10, seed 0

Evaluation. Following the PyTorch-Ignite recipe [2], images are resized to 299×299 (bilinear) and replicated to three channels. IS and FID use Inception-v3; FID uses the 2048-d pool3 features with the `pytorch-fid` weights via the Inception wrapper of the Ignite blog post. Samples are decoder means of $z \sim \mathcal{N}(0, I)$ (prior) or of z from a 10-component Gaussian mixture fitted to the training posterior means (GMM), compared with the 624 test images.

2 Results

Training. All terms fall quickly and flatten (Fig. 1). Final validation (train) values: MSE 0.0029 (0.0021) per pixel, KL 162 (163) nats, perceptual distance 1.81 (1.57). After the warm-up the KL settles at 139 nats and rises slowly to 163. After epoch 75 validation flattens while training keeps falling (overfitting); we keep the best validation checkpoint (epoch 137).

Preprint.

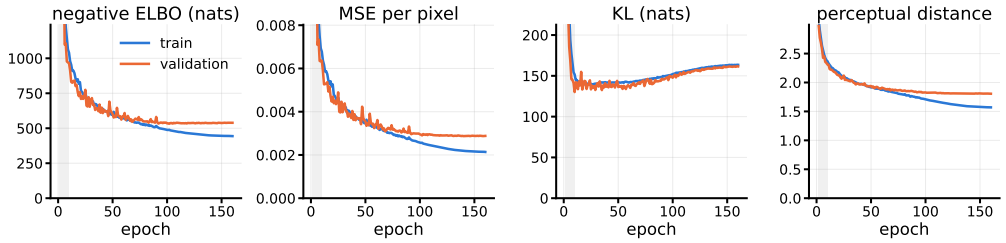
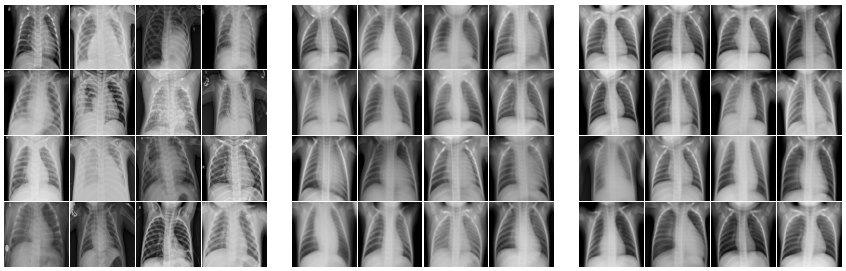


Figure 1: Curves per image: negative ELBO and MSE (without the Gaussian constant), KL and unweighted perceptual distance; shaded: KL warm-up, epoch 1 omitted.



(a) Real training images (b) Prior samples (c) GMM samples

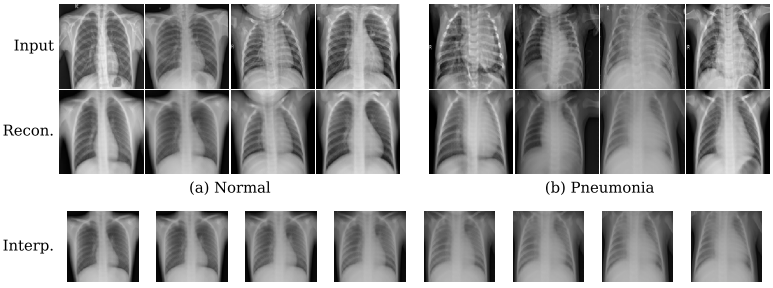
Figure 2: (a) Random real training images. (b) Prior samples, decoder mean of $z \sim \mathcal{N}(0, I)$. (c) Samples with z from the GMM fitted to the latent codes. For (b) and (c), 16 of 64 random samples were selected by eye; all metrics use unselected samples.

Table 2: Final model against the 624 test images; mean $\pm$ std over 10 (prior) and 6 (GMM) draws. Reconstruction MSE 0.0031; IS of the test images 2.17.

	FID ↓	IS ↑
Prior samples	126.0 ± 1.7	1.64 ± 0.02
GMM samples	116.4 ± 2.0	1.87 ± 0.02
Reconstructions	114.3	1.79
Real training images	26.4	2.17

Table 3: Validation FID (537 images, one draw; real images: 26.6–27.3). Each row adds to the row above; 60 epochs except the last three.

Setup	Recon.	Prior
Pixel loss, $\sigma = 0.1$	270.4	331.6
$\sigma = 0.25$	263.8	292.8
+ residual blocks	232.4	234.2
+ perceptual ($\lambda = 480$), shifts	224.2	212.6
+ 160 epochs	154.5	151.2
+ $\lambda = 1440$	143.4	141.4
+ upsampling decoder (final)	117.7	123.2



(a) Normal (b) Pneumonia

(c) Latent interpolation from a normal (left) to a pneumonia (right) test image

Figure 3: Evenly spaced test images and reconstructions of the (a) normal and (b) pneumonia class. (c) Decoded linear interpolation between the posterior means of a normal and a pneumonia test image.

Generated images. Prior samples reach FID 126.0 and IS 1.64 (Table 2), against FID 26.4 for real training images. They show coherent chests with ribs, heart, diaphragm and clavicles (Fig. 2). GMM samples are better (116.4), so the prior covers some regions the decoder rarely saw. All images are smoother than real X-rays: fine vessels, wires and side markers are missing.

Reconstructions and latent space. Reconstructions of both classes keep anatomy and ribs but smooth out the fine opacities of pneumonia; interpolations change smoothly (Fig. 3).

What made the images sharper. Every step in Table 3 helped; the perceptual loss brings back the ribs. A sub-pixel decoder helped less (prior FID 135). Not helpful were a much larger decoder without perceptual loss, training at 256×256 and, in an early setup, a learning rate of $2 \cdot 10^{-4}$.

Limitations. Images stay smoother than real X-rays, comparisons use one seed, and FID and IS say little about medical validity.

References

[1] Daniel S. Kermany et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131, 2018.

[2] Victor Fomin, Jeff Anmol, Sylvain Desroziers, Jeremy Kriss, and Alykhan Tejani. PyTorch-Ignite: Training and evaluating neural networks flexibly and transparently. `https://github.com/pytorch/ignite`, 2020.